

MARIO D. A. QUIROGA

FULL STACK ENGINEER

CONTACT

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PROFILE

Full Stack Engineer with 5+ years building and scaling production systems for high-traffic conversational AI platforms. Technical Leader of the integrations team of **Boti**, the official WhatsApp chatbot of the Buenos Aires City Government (~5M conversations/month). Strong backend in Java 21, Node.js and NestJS microservices, frontend with React + TypeScript, hands-on Google Cloud and production LLM work: OpenAI integrations, RAG pipelines and AI agents.

TECH STACK

Backend
- Java 21
- Node.js / NestJS
- Python (Flask)
- PHP / Laravel

Frontend
- React + TypeScript
- Vite / React Testing Library

Cloud & DevOps
- Google Cloud Platform (GKE, Cloud Run, Firebase)
- Docker · Git / GitLab CI

Data
- PostgreSQL / MySQL
- BigQuery / Firestore

Architecture
- RabbitMQ / event-driven
- REST & SOAP integrations

AI / LLM
- OpenAI API / LangGraph
- RAG & vector search (pgvector)
- TensorFlow / Keras (model training)

Ways of working
- Scrum · TDD · code review · Jira

Languages
- Spanish (native) · English (advanced)

EDUCATION

Bachelor's Degree in Information Systems — National University of Luján (UNLu), 2009–2022 · 41 of 42 courses completed, pending thesis defence.

University Programming Analyst — UNLu, 2019 · intermediate degree obtained.

Extra courses – UNLu: Enterprise applications with Spring & Hibernate applying TDD (2019); network programming in the TCP/IP stack (2018).

WORK EXPERIENCE

Linepros
07/2024 – PRESENT

Full Stack Engineer

- Leading the migration of a sports-betting management platform from Vanilla JavaScript to React + TypeScript, improving maintainability and delivery speed.
- Re-architected the **bet validation flow** from a per-partner synchronous process — one deployment per partner, **100+ running in parallel** — into a single asynchronous event-driven pipeline (**NestJS** + **RabbitMQ**) serving **100+ partners across 4 betting platforms**, where new partners are onboarded by configuration instead of a dedicated deployment.
- Designing and shipping internal **AI agents** with **LangGraph** and **RAG** pipelines (PostgreSQL + pgvector) to automate internal operational processes.
- Delivering end-to-end features across frontend and backend; built PHP services with optimised MySQL queries and stored procedures serving 100+ betting partners.
- Introduced component testing with React Testing Library, reducing regressions in migrated screens.
- Acting as technical escalation point for the Customer Service team, diagnosing and fixing partner-reported issues.

Technologies: TypeScript, React, NestJS, Node.js, RabbitMQ, LangGraph, pgvector, PostgreSQL, PHP, MySQL, GitLab.

Botmaker
11/2021 – 07/2024

Technical Leader – Boti Integrations Team | 01/2023 – 07/2024

- Led the integrations squad for **Boti**, the official chatbot of the Buenos Aires City Government (~5M monthly conversations; 58M+ in 2022), owning technical decisions, estimations and delivery for the client.
- Designed and delivered REST and SOAP integrations with multiple government systems (certificates, appointments, claims), enabling citizens to complete procedures end-to-end inside WhatsApp.
- Translated government requirements into technical specifications and release plans, coordinating and mentoring a team of 4 engineers through code review and pairing while sustaining 24/7 availability of a critical public service.

Full Stack Engineer | 11/2021 – 12/2022

- Designed, developed and maintained product features for a multi-tenant conversational platform using React on the frontend and Java 21 on the backend.
- Deployed and scaled services on Google Cloud (Kubernetes, Cloud Run, Cloud Functions), using BigQuery and Firestore for analytics and real-time data.
- As a member of the **AI Research** team, trained and evaluated models on historical conversation data and integrated the OpenAI API into the bot core to improve intent recognition and response accuracy.

Technologies: Java 21, React, Node.js, Python (Flask), GCP (Kubernetes, Cloud Run, Cloud Functions, BigQuery, Firestore, Firebase), PostgreSQL, OpenAI API, Scrum.

National University of Luján (UNLu)
03/2019 – 05/2024

Software Developer | 06/2021 – 04/2024

- Analysed, designed and built the Teaching Staff Management System for the university's Technology Department, replacing manual spreadsheet workflows across the department.

Technologies: Laravel, PHP, Blade, MySQL, Docker.

Teaching Assistant | 05/2021 – 05/2024

- Taught Introduction to Programming, Programming 1 and Graph & Network Mining in the Information Systems degree.
- Designed practical assignments and automated delivery and grading with Git and GitHub Classroom.

Technologies: Python, C, Git, GitHub Classroom.